

TT Compression of Attention in a Molecular Transformer

Can TT compress a molecular Transformer
without losing predictive accuracy?

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PhD at the Delft University of Technology, research on:

- Tensor Networks for Large Scale Kernel Machines
- Connections between Tensor Networks, Kernel Machines and Deep Nets
- Kernel extensions of MLSVD

Data Scientist at Equinor, working on:

- Omnia.Prevent: Anomaly Detection at scale
- FiberPredict: Anomaly Detection in wells using Fiber Optics
- Ensemble Modelling

Data and MAT

Target: aqueous solubility, $\log S$

1,128

raw records
in SMILES notation

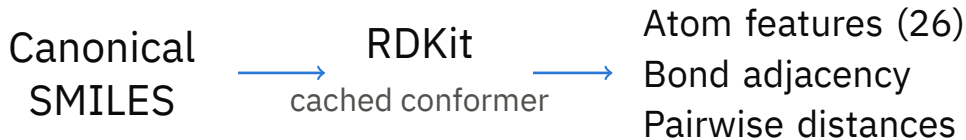


1,117

unique molecules
canonical SMILES

Duplicate labels averaged before splitting.

11 duplicate groups, six with conflicting labels.



80/10/10 random split, redrawn for each seed

893

Train

train-only target
scaling

112

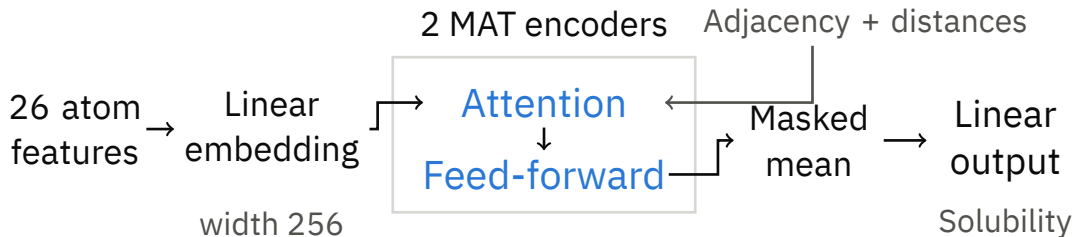
Validation

validation-only
selection

112

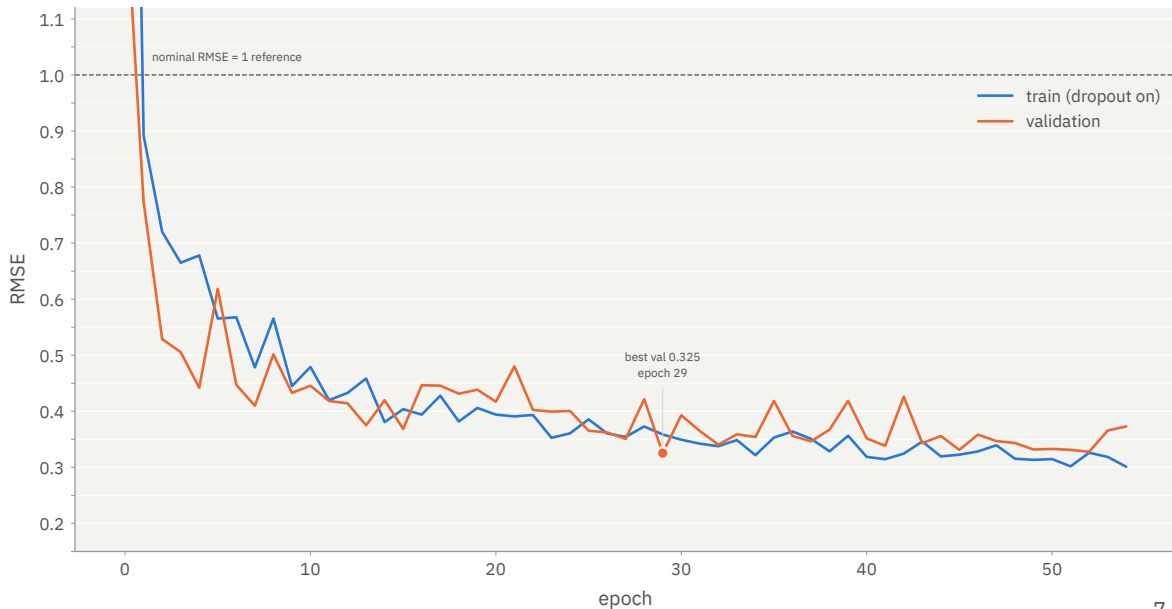
Test

one final test
evaluation



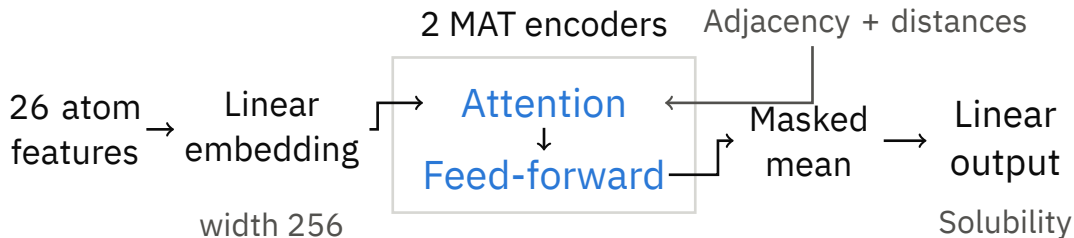
Dense Baseline

Dense MAT training curve (seed 0)



TT Compression

TT replaces 8 attention and 4 feed-forward weight matrices

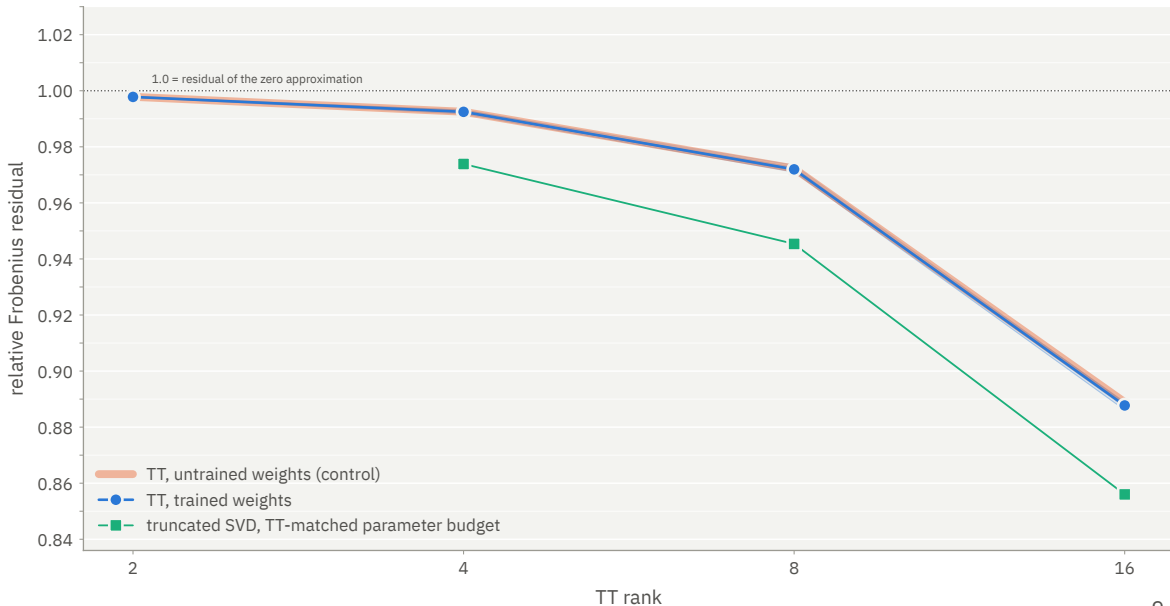


TT replaces 12 square weight matrices

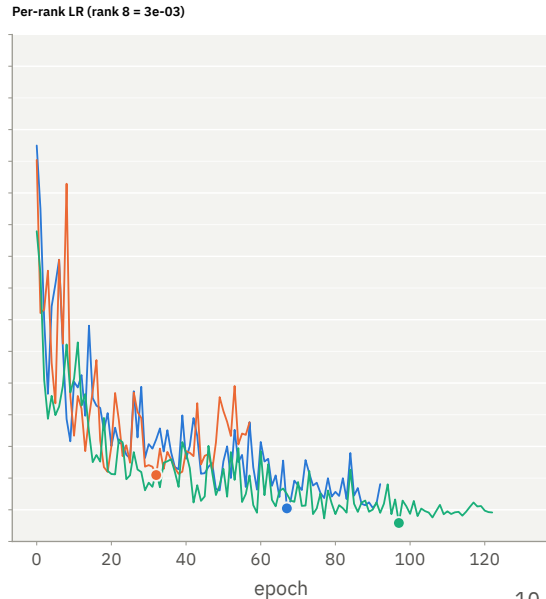
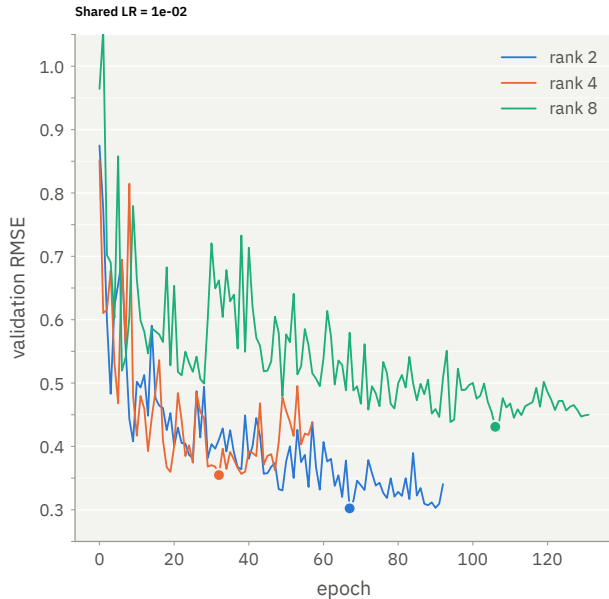
Attention projections + feed-forward layers, each 256×256

Embedding, normalization and output are unchanged.

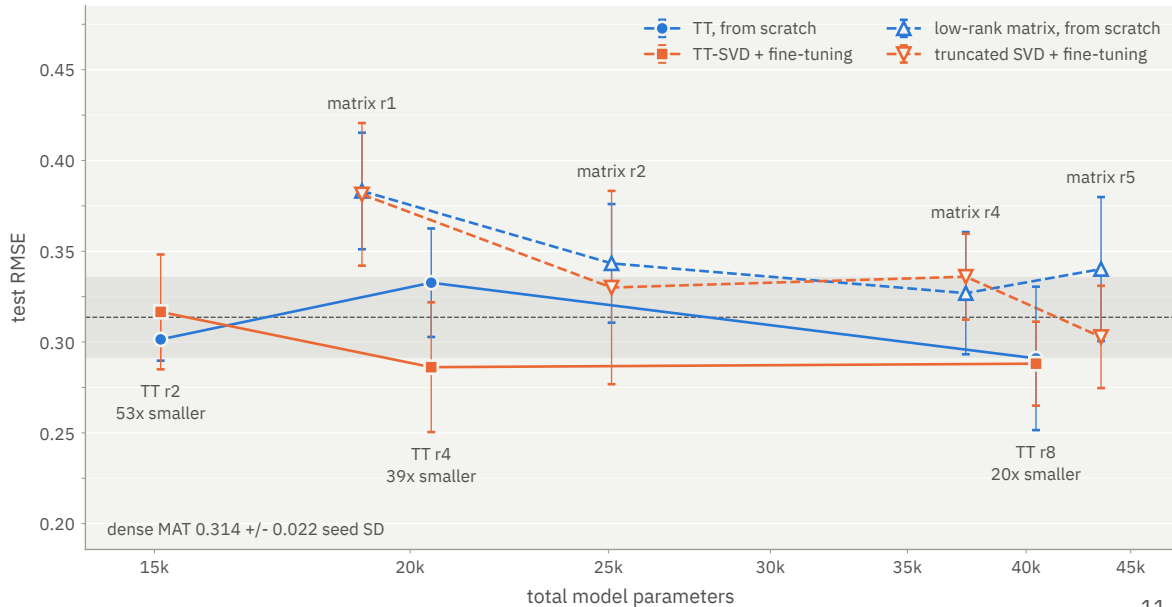
TT-SVD poorly reconstructs trained 256x256 MAT weights



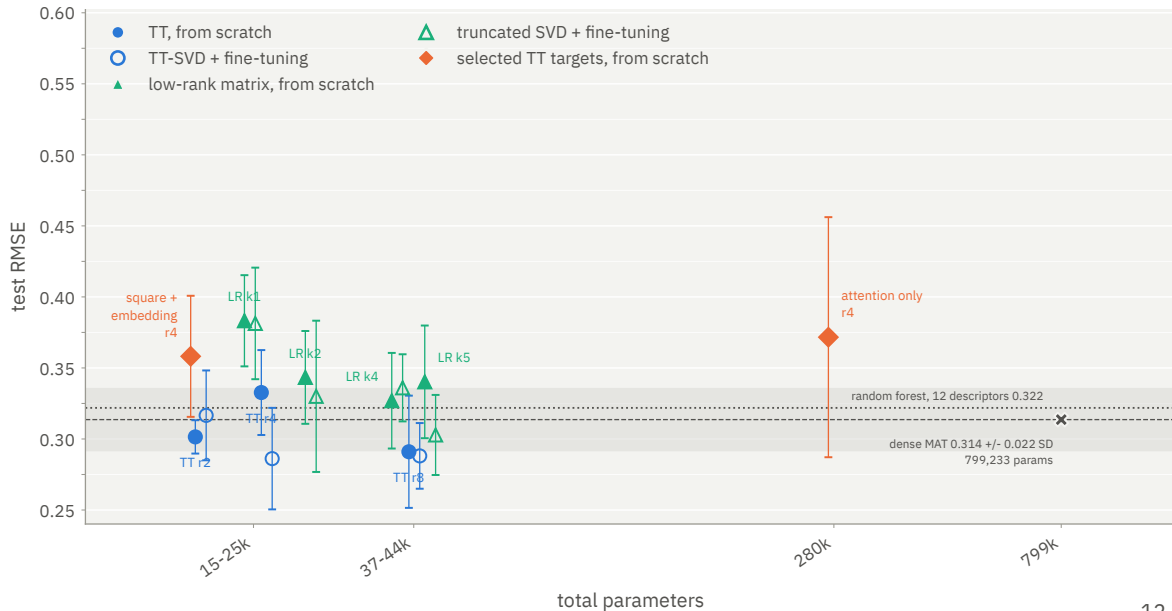
Rank is sensitive to learning rate (seed 0, rank 8)



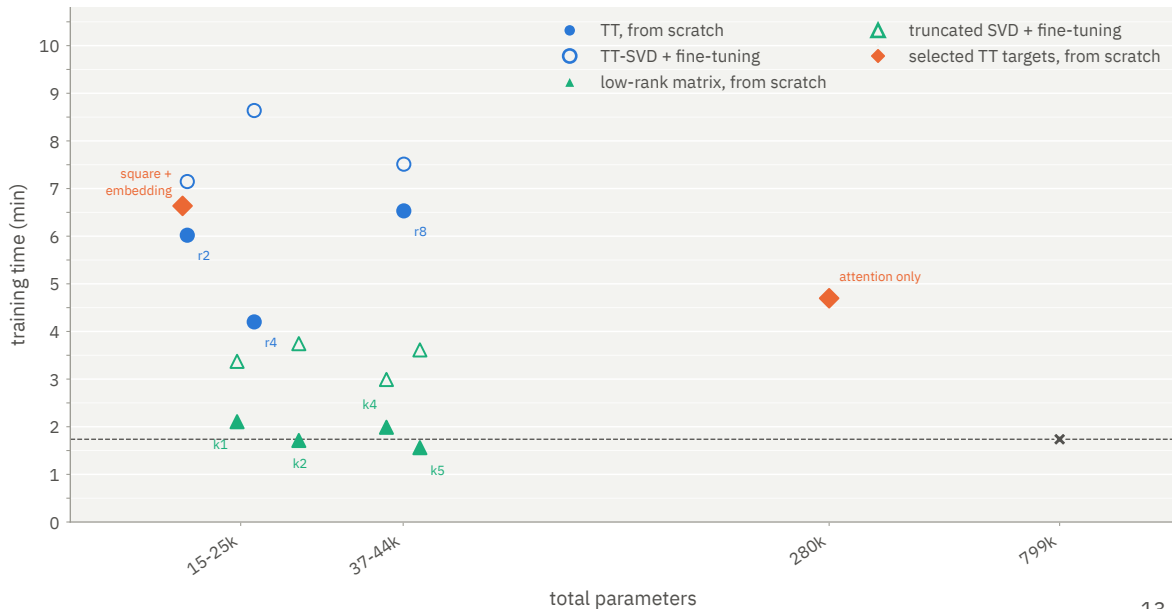
Test RMSE by parameter count



Test RMSE versus total model parameters



Training time versus total model parameters



Interpretation

MODEL RMSE IN PHYSICAL UNITS

MAT-standardized $\times 2.0955$

This study: $0.314 \rightarrow 0.66 \log S$

MAT paper: $0.285 \rightarrow 0.60 \log S$

MAT RMSE close to irreducible error.

Is there room for improvement?

LITERATURE-DATA UNCERTAINTY

$0.6-0.7 \log S$

Palmer and Mitchell (2014)

Measurement uncertainty
on solubility data
in compiled datasets.

6 conflicting pairs: $0.31 \log S$ conditional per-measurement SD
(under i.i.d. assumption)

However the sampling criteria behind these pairs not known.

TT compresses most parameters but does not demonstrate chemical structure

PARAMETER STORAGE

Attention projections
+ feed-forward layers

98.8%

of parameters in this MAT

CHEMICAL STRUCTURE-RELATED COMPRESSION

Atom-feature embedding

Chemically organized features
versus randomized organization

Small storage contribution,
chemically identified inputs.

Compression alone does not demonstrate chemical structure.

Things I would
have liked to try

First reproduce MAT, then compare factorizations

Paper reference: RMSE 0.285. Match the model and evaluation protocol.

Folding per axis	Tensor train	Tensor ring
Radix-4, 4^4	Current TT	Radix-4 TR
Binary, 2^8	Binary TT (QTT)	Binary TR

Same hyperparameters as MAT, same folds.

Matched parameters, not matched ranks. Test all four combinations.

Proposed ESOL comparison controls splits, initialization and tuning

SPLITS AND SEEDS

Six paper splits

Three initializations per split

Paired seeds.

VALIDATION-ONLY TUNING

Rank/budget, learning rate,
weight decay and dropout

Equal search budget per condition.

ILLUSTRATIVE COMPUTE BUDGET

7 conditions $\times$ 6 splits $\times$ 3 initializations = 126 final fits

Assumed 10 min/fit on one A100 80 GB + 20% contingency

$126 \times 10/60 \times 1.20 = 25.2$ **GPU-hours**

One parameter budget and training path. Runtime is not measured.

Baseline reproduction and tuning are additional, estimate after a pilot.

Stop condition: no useful trade-off over low rank or smaller dense MAT.

Conclusions

- 20–53 $\times$ fewer parameters, with no detected predictive loss.
- TT training was slower on CPU.
- ESOL has conflicting labels and is very likely noisy.
- Compressing the model has little business value.

